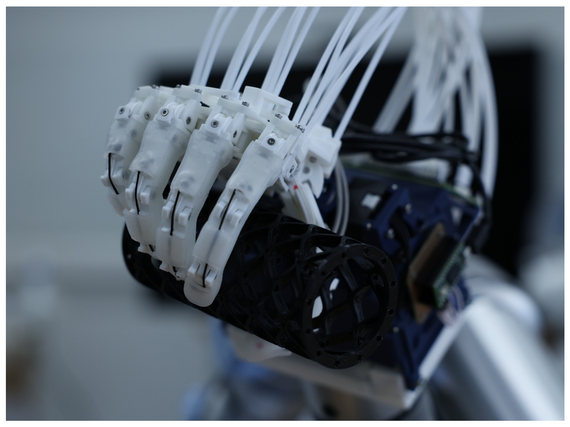
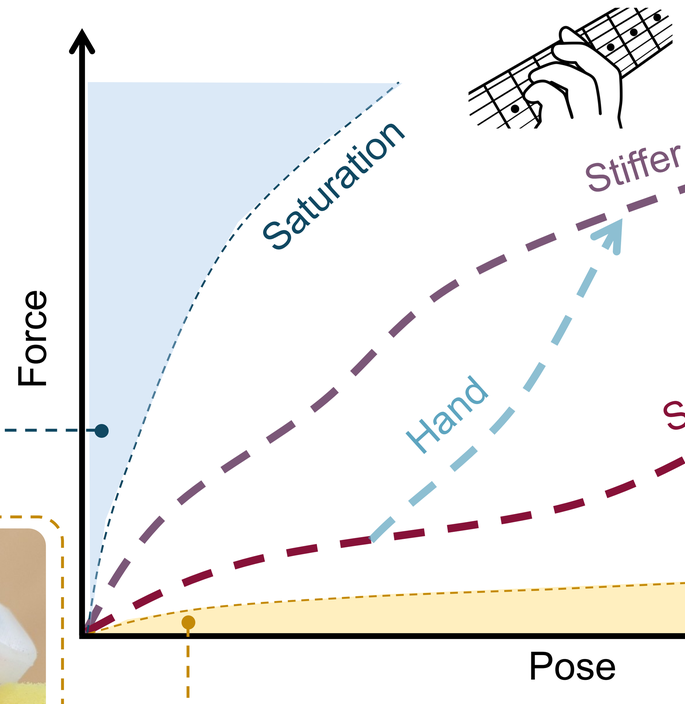
a

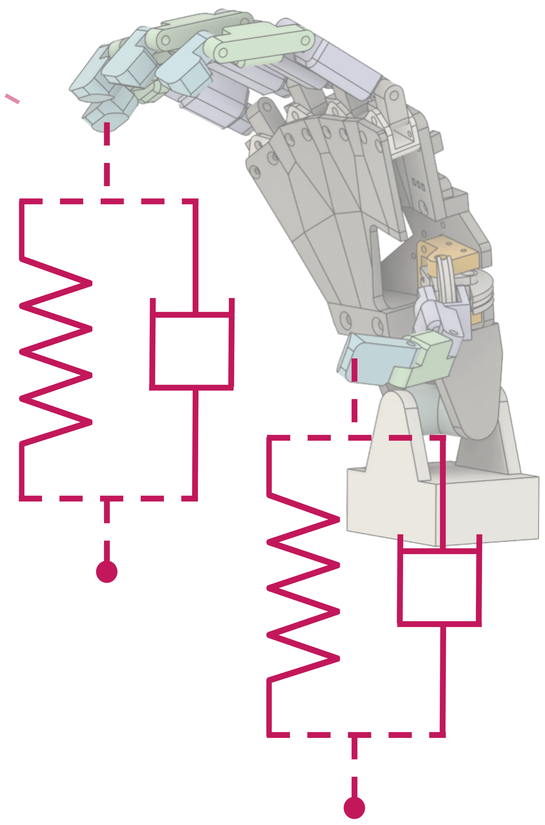
Bio-inspired Distributed Compliance:
Physical limits on safety and stability

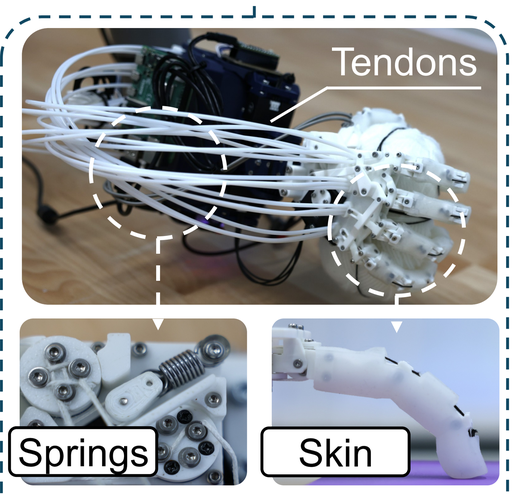
VMC Augmented Hand

Virtual Mechanisms:
Render different hand properties

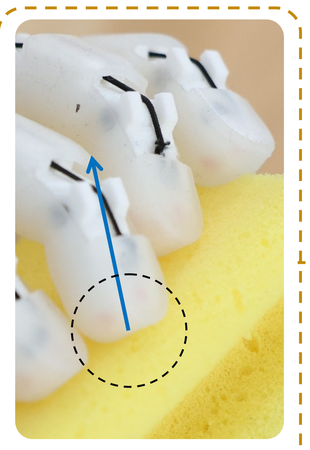




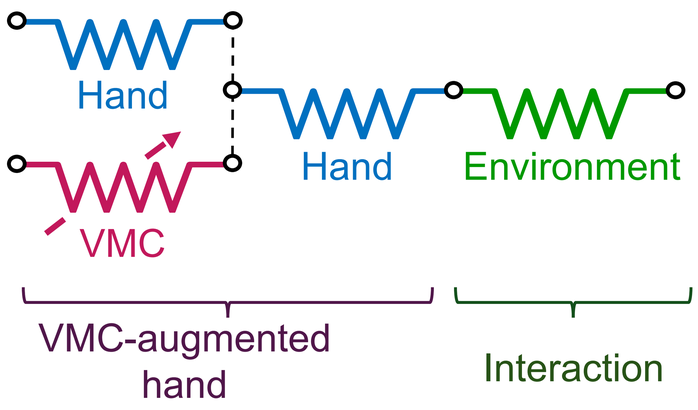




Series elastic actuation: Safety upper bound



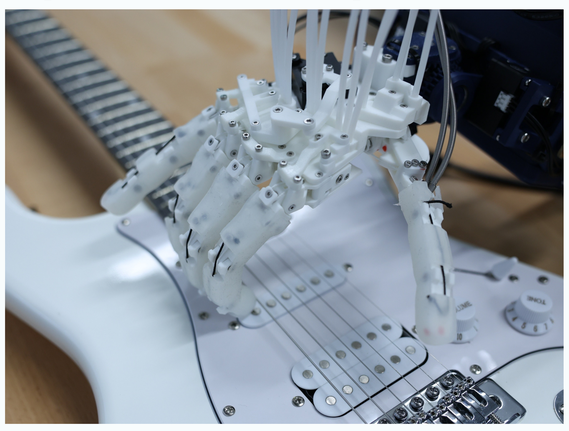
Skin & Tendon friction: Contact stability

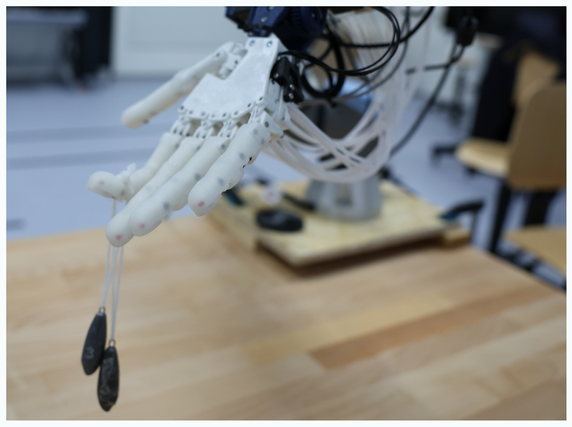


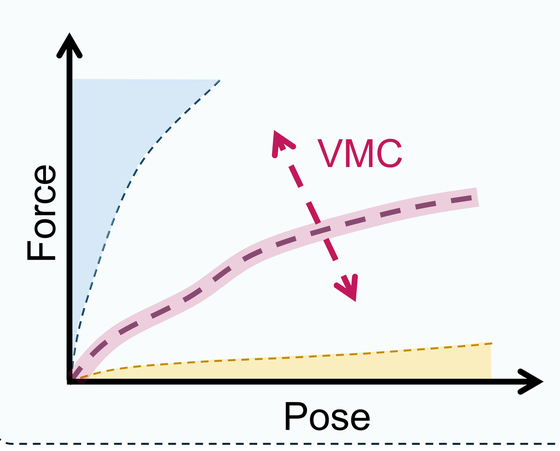
Virtual mechanics

b

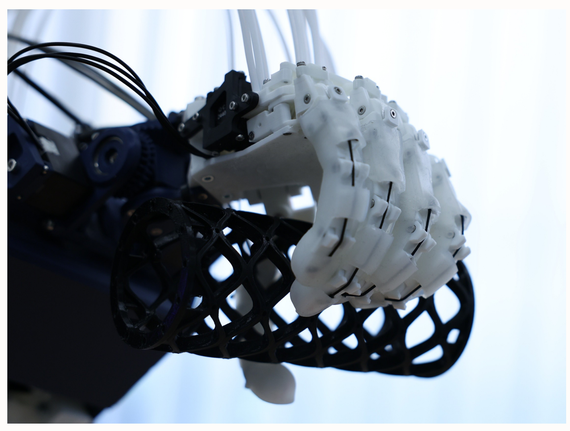
Programmable passive compliance

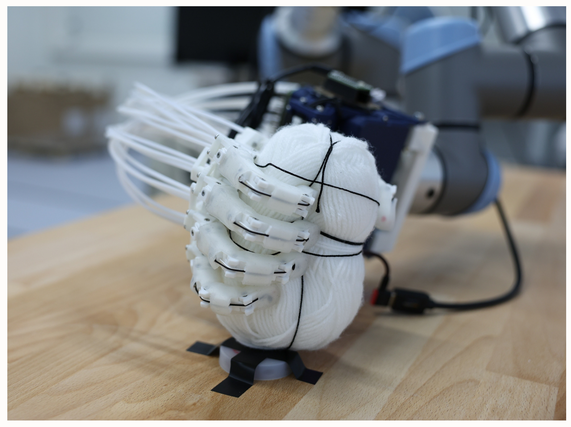


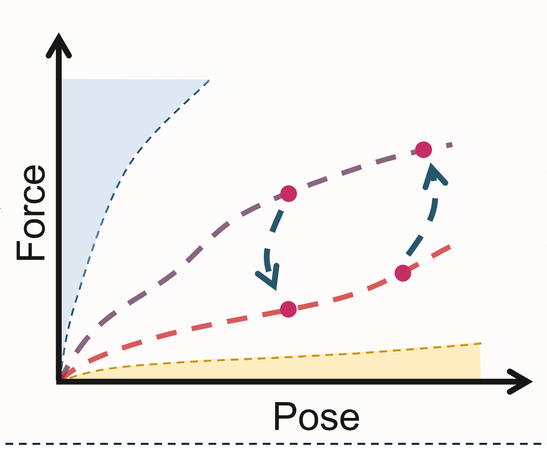




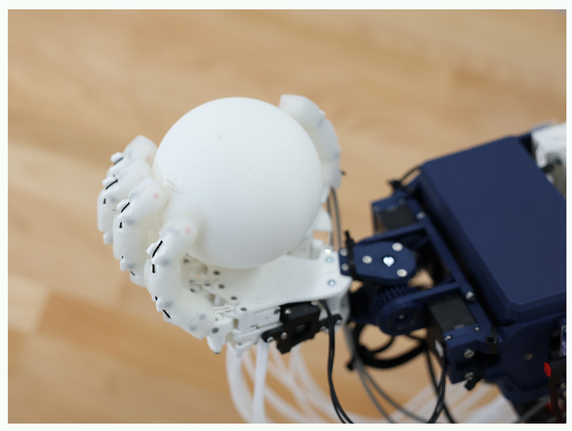
Online stiffness modulation

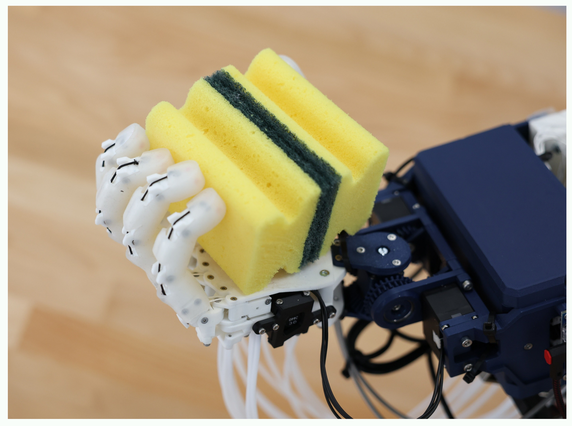


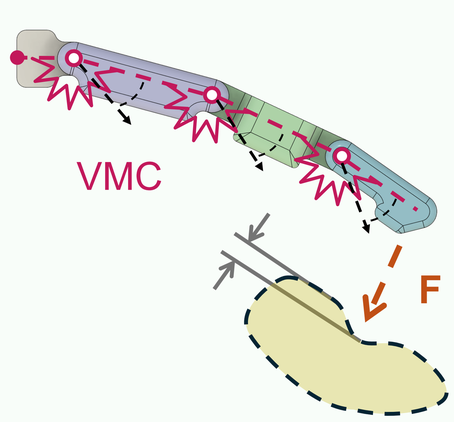
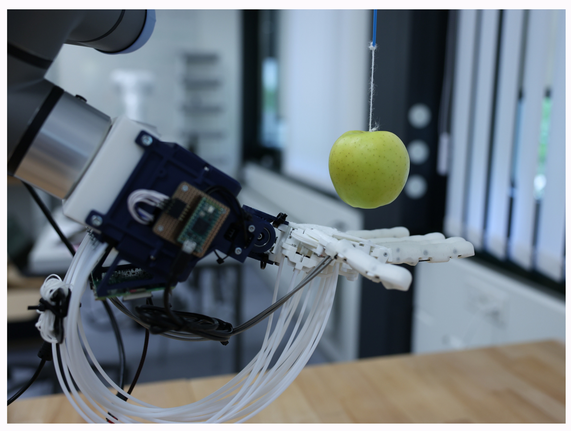
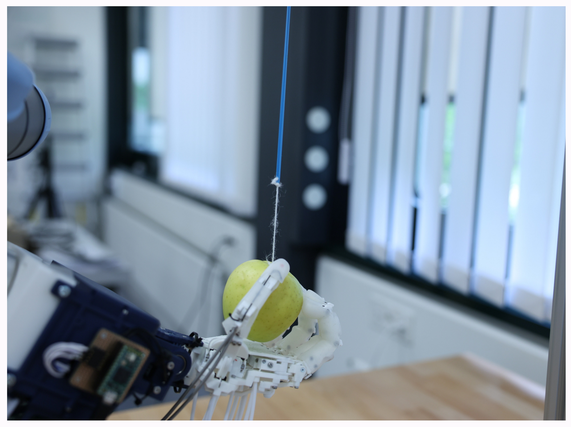
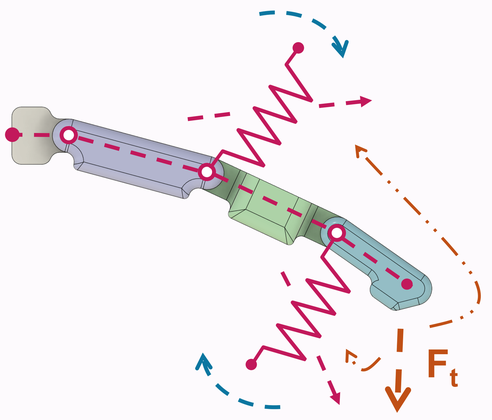




Sensorless exteroception







Optimized force control